

An energy company is considering an investment in wind-power production. To determine whether this investment is economically feasible, the company seeks to estimate the expected revenue of a wind-power producer operating in a competitive spot market.

Consider a day-ahead spot market with five electricity producers. Three producers generate electricity from conventional sources, such as coal, whereas the remaining two rely on solar and wind power, respectively. The market is cleared once each day for 24 hourly trading intervals of the following day. For each interval, every producer submits a bid that specifies the electricity quantity it is willing to supply (MWh), together with the minimum price it is willing to accept ($\$/\text{MWh}$). The market operator then applies a merit-order clearing mechanism to determine both the market-clearing price and the dispatch schedule, subject to the requirement that total electricity demand be satisfied. If a producer's bid is accepted, the producer is obligated to supply the nominated quantity and is remunerated at the market-clearing price. If the bid is not accepted, the producer receives no revenue for that interval.

The energy company intends to develop a model to estimate the expected revenue of the wind-power producer on any future day by determining the hourly bid quantities and prices under uncertainty. On this basis, the model can subsequently be extended to examine outcomes across a range of short-term and long-term scenarios, thereby supporting an assessment of the investment's economic viability.

Please formalise this problem as an agent-based model, and specify the key components, including time scales, agents, state variables, control variables, processes, objectives, constants, uncertain parameters and stochastic variables. At this stage, the model should be limited to simulating single-day market operations.

Show less

Agent-Based Model: Wind-Power Producer in a Day-Ahead Spot Market

1. Overview

The model simulates a single day of operations in a day-ahead electricity spot market. Five competing producers submit bids for each of 24 hourly intervals. A merit-order clearing mechanism determines the market-clearing price and dispatch schedule for each interval. The objective is to estimate the expected daily revenue of the wind-power producer.

2. Time Scales

Scale	Description
Planning horizon	A single future day, indexed d
Trading intervals	$t \in \mathcal{T} = \{1, 2, \dots, 24\}$, each representing one clock hour
Clearing event	One market-clearing per interval t , executed simultaneously for all t before the day begins

All bids are submitted once, before the day, in a single sealed-bid round. There is no intra-day re-bidding.

3. Agents

The model contains **five producer agents** and one **market operator**.

3.1 Producer Agents

Agent	Index	Technology	Role in model
Conventional 1-3	$i \in \{C_1, C_2, C_3\}$	Coal	Strategic or price-taking competitors
Solar producer	$i = S$	Solar PV	Competitor; output weather-dependent
Wind producer	$i = W$	Wind	Focal agent ; revenue to be estimated

Each producer is an autonomous decision-maker that observes its own costs and resource forecasts, forms beliefs about competitor behaviour, and submits a bid vector for the day.

3.2 Market Operator (Non-Strategic)

A passive clearing agent that collects all bids, applies the merit-order algorithm, and returns prices and dispatch decisions. It has no objective of its own and exercises no discretion.

4. State Variables

State variables describe the condition of the system at each interval t before decisions are executed.

Symbol	Domain	Description
D_t	$\mathbb{R}_{>0}$	Aggregate electricity demand (MWh) in interval t
$\tilde{v}_t$	$[0, v^{\max}]$	Realised wind speed (m/s) at the wind site in interval t
$\tilde{r}_t$	$[0, r^{\max}]$	Realised solar irradiance (W/m ²) at the solar site in interval t
f_t^W	$[0, \bar{Q}^W]$	Available wind generation capacity (MWh); derived from $\tilde{v}_t$ via a power curve
f_t^S	$[0, \bar{Q}^S]$	Available solar generation capacity (MWh); derived from $\tilde{r}_t$
λ_t	$\mathbb{R}_{\geq 0}$	Market-clearing price (\$/MWh) in interval t ; an outcome, not set by any agent

5. Control Variables

Control variables are the decision variables each producer agent chooses when submitting its bid.

For producer i and interval t :

Symbol	Domain	Description
q_t^i	$[0, \bar{Q}^i]$	Bid quantity (MWh)
p_t^i	$[\underline{p}, \bar{p}]$	Bid price (\$/MWh); minimum acceptable remuneration

The **wind producer's** control vector for the day is:

$$\mathbf{u}^W = \{(q_t^W, p_t^W)\}_{t=1}^{24}$$

Feasibility constraints on the wind producer's bid:

$$0 \leq q_t^W \leq f_t^W \quad \forall t \quad (1)$$

$$\underline{p} \leq p_t^W \leq \bar{p} \quad \forall t \quad (2)$$

Constraint (1) ensures the producer cannot bid more than its available wind resource. A common strategy for renewable producers with near-zero marginal cost is to set $p_t^W = 0$ and $q_t^W = f_t^W$ (price-taking, full-output bid).

6. Processes

6.1 Wind and Solar Resource Realisation

Before bidding, each renewable producer receives a **forecast** of its available capacity. The actual available capacity is not known with certainty at bid time.

$$f_t^W = \mathcal{P}(\tilde{v}_t) \quad (3)$$

where $\mathcal{P}(\cdot)$ is the wind turbine power curve (a deterministic, non-linear mapping). Similarly, $f_t^S = \eta^S \cdot A^S \cdot \tilde{r}_t$ for the solar producer.

6.2 Bid Formation

Each producer i submits (q_t^i, p_t^i) for all $t \in \mathcal{T}$. Bids are sealed; no producer observes competitors' bids before submission.

6.3 Merit-Order Market Clearing

For each interval t , the market operator:

1. Ranks all accepted bids in ascending order of bid price p_t^i .
2. Cumulates quantities along this supply stack until total supply meets demand D_t .
3. Sets the **market-clearing price** λ_t equal to the bid price of the **marginal (last accepted) producer**.
4. All producers whose bid price $p_t^i \leq \lambda_t$ and whose quantity fits within the cleared stack are **dispatched** and receive λ_t .

Formally, let σ_t denote the merit-order ranking. The clearing price is:

$$\lambda_t = p_t^{\sigma_t(k^*)}, \quad k^* = \min \left\{ k : \sum_{j=1}^k q_t^{\sigma_t(j)} \geq D_t \right\} \quad (4)$$

Producer i is dispatched if $p_t^i \leq \lambda_t$ and it falls within the accepted stack.

6.4 Revenue Calculation

The wind producer's revenue for interval t is:

$$R_t^W = \lambda_t \cdot q_t^W \cdot \mathbf{1}[p_t^W \leq \lambda_t] \quad (5)$$

Total daily revenue:

$$R^W = \sum_{t=1}^{24} R_t^W \quad (6)$$

7. Objectives

Agent	Objective
Wind producer (focal)	Maximise expected daily revenue: $\max_{\mathbf{u}^W} \mathbb{E}[R^W]$
Conventional producers	Maximise profit (revenue minus fuel and operating costs)
Solar producer	Maximise expected revenue (analogous to wind)
Market operator	Minimise cost of supply while satisfying demand (implicit in merit-order)

At the single-day simulation stage, the wind producer's objective can be treated as **revenue estimation** (passive), with optimisation reserved for the model extension.

8. Constants and Fixed Parameters

These are assumed known and invariant throughout the simulation.

$$\left| \begin{array}{c} R^{\wedge}W = \sum_t R^{\wedge}W_t \\ \hline \end{array} \right| \quad \text{(daily total)}$$

11. Key Modelling Assumptions (Single-Day Scope)

1. **Perfect competition / price-taking** among renewable producers: wind and solar bid at zero marginal cost with full available output.
2. **Conventional producers bid at marginal cost**, meaning no strategic mark-up, unless a strategic extension is added.
3. **No transmission constraints**: the market is a single-node (copper-plate) network.
4. **No storage or ramp constraints**: each interval is cleared independently.
5. **Demand is inelastic**: D_t is exogenous and must be fully satisfied.
6. **Renewable producers are not penalised** for under-delivery relative to bid quantity (no imbalance settlement at this stage).
7. **Stochastic variables are independent across agents** but may be temporally correlated across intervals t for each agent.

These assumptions can be relaxed progressively in the model extension to support the longer-term investment assessment.

You can assume that the market demand follows a normal distribution with constant mean and variance. Please refine your specification accordingly and present the revised section.

9. Uncertain Parameters and Stochastic Variables (Revised)

These are not known at bid-submission time and are modelled probabilistically.

Symbol	Type	Description	Model
$\tilde{v}_t$	Stochastic	Wind speed at interval t	Weibull distribution; temporally correlated (AR or Markov chain)
$\tilde{r}_t$	Stochastic	Solar irradiance at interval t	Beta or truncated normal; correlated with time-of-day and season
$\tilde{D}_t$	Stochastic	Aggregate electricity demand at interval t	Normal distribution with constant mean and variance (see below)
$\tilde{p}_t^{C_k}$	Uncertain	Bid prices of conventional producers	Fixed at marginal cost c^{C_k} under price-taking assumption, or drawn from a distribution under strategic bidding
$\tilde{q}_t^{C_k}$	Uncertain	Bid quantities of conventional producers	Typically $= \bar{Q}^{C_k}$ under full-commitment assumption
$\tilde{f}_t^W$	Derived stochastic	Wind available capacity; function of $\tilde{v}_t$ via (3)	Inherited from wind speed model
$\tilde{f}_t^S$	Derived stochastic	Solar available capacity; function of $\tilde{r}_t$	Inherited from irradiance model

Demand Specification

Aggregate demand in each interval t is modelled as an independent normal random variable with constant mean μ_D and variance σ_D^2 :

$$\tilde{D}_t \sim \mathcal{N}(\mu_D, \sigma_D^2) \quad \forall t \in \mathcal{T} \quad (7)$$

with $\tilde{D}_t$ independent across intervals. Two additional parameters therefore join the set of constants:

Symbol	Description	Unit
μ_D	Mean aggregate demand, common to all intervals	MWh
σ_D^2	Variance of aggregate demand, common to all intervals	MWh ²

****Implicit constraints.**** Since physical demand must be strictly positive, realisations $\tilde{D}_t \leq 0$ are treated as infeasible. Provided the coefficient of variation σ_D/μ_D is small — a reasonable assumption for aggregate grid demand — the probability of such realisations is negligible and can be ignored without material effect on the model. If tighter control is required, a truncated normal $\mathcal{N}_{(0,\infty)}(\mu_D, \sigma_D^2)$ may be substituted at the cost of minor additional complexity.

Implications for market clearing. Because μ_D and σ_D^2 are constant across intervals, the model abstracts away intra-day demand variation (e.g. morning peaks and overnight troughs). This is consistent with a single-day simulation focused on revenue estimation under uncertainty, rather than on the detailed shape of the daily load profile. The assumption can be relaxed in a model extension by replacing the constant parameters with interval-specific means and variances, $\mu_{D,t}$ and $\sigma_{D,t}^2$, to capture diurnal demand patterns.

You can also make the following assumptions regarding the behavioural logic of the producers: (1) conventional producers have stable bid quantities, with bid prices following normal distributions with uncertain means and variances; (2) the solar-power producer's bid quantity follows $\max(0, a+b\cos(\frac{2\pi t}{24}))$, with bid prices following a normal distribution with uncertain mean and variance; and (3) actual wind-power production is stochastic, following a normal distribution with constant mean and variance. Please refine your specification accordingly and present the revised section.

9. Uncertain Parameters and Stochastic Variables (Revised)

These are not known at bid-submission time and are modelled probabilistically. The behavioural logic of each producer class is specified below.

9.1 Demand

Aggregate demand in each interval t is an independent normal random variable with constant mean and variance, unchanged from the previous revision:

$$\tilde{D}_t \sim \mathcal{N}(\mu_D, \sigma_D^2) \quad \forall t \in \mathcal{T} \quad (7)$$

Symbol	Description	Unit
μ_D	Mean aggregate demand, common to all intervals	MWh
σ_D^2	Variance of aggregate demand, common to all intervals	MWh ²

9.2 Conventional Producers

Each conventional producer $k \in \{C_1, C_2, C_3\}$ commits its full installed capacity in every interval, with bid prices drawn from a normal distribution whose mean and variance are themselves uncertain.

Bid quantities are stable and fixed at installed capacity:

$$q_t^{C_k} = \bar{Q}^{C_k} \quad \forall t \in \mathcal{T} \quad (8)$$

Bid prices are drawn independently across intervals from a normal distribution:

$$\tilde{p}_t^{C_k} \sim \mathcal{N}(\mu_p^{C_k}, \sigma_p^{2,C_k}) \quad \forall t \in \mathcal{T} \quad (9)$$

where the distributional parameters $\mu_p^{C_k}$ and σ_p^{2,C_k} are not known with certainty and are themselves treated as uncertain parameters, to be estimated or drawn from prior distributions in simulation. As with demand, realisations $\tilde{p}_t^{C_k} < \underline{p}$ or $\tilde{p}_t^{C_k} > \bar{p}$ are truncated to the market price bounds $[\underline{p}, \bar{p}]$.

Symbol	Description	Unit
$\bar{Q}^{C_k}$	Installed (bid) capacity of conventional producer k	MWh
$\mu_p^{C_k}$	Uncertain mean bid price of conventional producer k	\$/MWh
σ_p^{2,C_k}	Uncertain variance of bid price of conventional producer k	(\$/MWh) ²

9.3 Solar Producer

The solar producer's available output follows a deterministic cosine profile — capturing the diurnal arc of solar irradiance — floored at zero to reflect the absence of generation at night.

Bid quantities are given by:

$$q_t^S = \max\left(0, a + b \cos\left(\frac{2\pi t}{24}\right)\right) \quad \forall t \in \mathcal{T} \quad (10)$$

where $a \in \mathbb{R}$ is the vertical offset and $b > 0$ is the amplitude. The cosine term attains its maximum at $t = 0$ (midnight) and its minimum at $t = 12$ (noon) under standard indexing; if $t = 1$ denotes the first hour of the day, the peak shifts accordingly. The floor at zero ensures no negative generation is bid. Note that $q_t^S = 0$ whenever $a + b \cos(2\pi t/24) \leq 0$, which defines the overnight zero-output window.

Bid prices follow a normal distribution with uncertain mean and variance, independently across intervals:

$$\tilde{p}_t^S \sim \mathcal{N}(\mu_p^S, \sigma_p^{2,S}) \quad \forall t \in \mathcal{T} \quad (11)$$

Symbol	Description	Unit
a	Vertical offset of solar generation profile	MWh
b	Amplitude of solar generation profile	MWh
μ_p^S	Uncertain mean bid price of the solar producer	\$/MWh
$\sigma_p^{2,S}$	Uncertain variance of bid price of the solar producer	(\$/MWh) ²

9.4 Wind Producer

Actual wind-power production is stochastic, modelled as an independent normal random variable with constant mean and variance across all intervals. This abstracts away temporal wind patterns in favour of tractability at the single-day simulation stage.

Bid quantities are set equal to the expected wind output, i.e. the producer bids its mean available capacity:

$$q_t^W = \mu_q^W \quad \forall t \in \mathcal{T} \quad (12)$$

Actual production is stochastic and realised after the market clears:

$$\tilde{f}_t^W \sim \mathcal{N}(\mu_q^W, \sigma_q^{2,W}) \quad \forall t \in \mathcal{T} \quad (13)$$

As with demand, realisations $\tilde{f}_t^W < 0$ are physically infeasible; provided σ_q^W / μ_q^W is small, the probability of such realisations is negligible. A truncated normal $\mathcal{N}_{(0, \bar{Q}^W)}(\mu_q^W, \sigma_q^{2,W})$ may be substituted if required.

Bid prices are set to zero, consistent with the price-taking assumption for a near-zero-marginal-cost renewable producer:

$$p_t^W = 0 \quad \forall t \in \mathcal{T} \quad (14)$$

This ensures the wind producer is always dispatched when demand is sufficient, and the market-clearing price is determined entirely by the conventional and solar producers.

Symbol	Description	Unit
μ_q^W	Mean wind production (= bid quantity)	MWh
$\sigma_q^{2,W}$	Variance of actual wind production	MWh ²

9.5 Summary Table

Symbol	Applies to	Type	Description
$\tilde{D}_t$	Market	Stochastic	Demand $\sim \mathcal{N}(\mu_D, \sigma_D^2)$, i.i.d. across t
$q_t^{C_k}$	Conventional k	Constant	Fixed at $\bar{Q}^{C_k}$
$\tilde{p}_t^{C_k}$	Conventional k	Stochastic	Bid price $\sim \mathcal{N}(\mu_p^{C_k}, \sigma_p^{2,C_k})$, uncertain parameters
q_t^S	Solar	Deterministic	Cosine profile $\max(0, a + b \cos(2\pi t/24))$
$\tilde{p}_t^S$	Solar	Stochastic	Bid price $\sim \mathcal{N}(\mu_p^S, \sigma_p^{2,S})$, uncertain parameters
q_t^W	Wind	Constant	Fixed at μ_q^W (price-taking, full expected output)
$\tilde{f}_t^W$	Wind	Stochastic	Actual production $\sim \mathcal{N}(\mu_q^W, \sigma_q^{2,W})$, i.i.d. across t
p_t^W	Wind	Constant	Fixed at 0 \$/MWh

When multiple bids at the clearing price collectively exceed the residual demand, producers are dispatched in priority order: solar $\rightarrow$ wind $\rightarrow$ conventional, with the last accepted producer potentially dispatched only partially to exactly meet demand. Please refine your specification accordingly and present the revised section.

6. Processes (Revised Section 6.3)

6.3 Merit-Order Market Clearing

For each interval t , the market operator clears the market in two stages: price-based ranking followed by priority-based rationing at the margin.

Stage 1: Merit-Order Ranking

All bids are ranked in ascending order of bid price. Let σ_t denote this ranking, so that $p_t^{\sigma_t(1)} \leq p_t^{\sigma_t(2)} \leq \dots \leq p_t^{\sigma_t(5)}$. The market operator cumulates dispatched quantities along the stack until aggregate supply meets demand $\tilde{D}_t$. The **market-clearing price** is set at the bid price of the marginal producer:

$$\lambda_t = p_t^{\sigma_t(k^*)}, \quad k^* = \min \left\{ k : \sum_{j=1}^k q_t^{\sigma_t(j)} \geq \tilde{D}_t \right\} \quad (4)$$

All producers with bid price strictly below λ_t are **fully dispatched**. All producers with bid price strictly above λ_t are **not dispatched**. The set of producers bidding exactly at λ_t forms the **marginal cohort** $\mathcal{M}_t$:

$$\mathcal{M}_t = \{i : p_t^i = \lambda_t\} \quad (5)$$

Stage 2: Priority Dispatch Within the Marginal Cohort

When $|\mathcal{M}_t| > 1$ and the marginal cohort's collective capacity exceeds residual demand, a strict priority order governs dispatch. Define the **residual demand** at the margin as the

demand remaining after all infra-marginal producers (those with $p_t^i < \lambda_t$) have been fully dispatched:

$$\tilde{D}_t^{\text{res}} = \tilde{D}_t - \sum_{\{i: p_t^i < \lambda_t\}} q_t^i \quad (6)$$

Producers in $\mathcal{M}_t$ are dispatched sequentially according to the priority order:

$$S \succ W \succ C_1 \succ C_2 \succ C_3$$

Let $\pi = (\pi_1, \pi_2, \dots)$ denote the producers in $\mathcal{M}_t$ sorted by this priority. Each producer π_j is dispatched as follows:

$$d_t^{\pi_j} = \min \left(q_t^{\pi_j}, \max \left(0, \tilde{D}_t^{\text{res}} - \sum_{l < j} d_t^{\pi_l} \right) \right) \quad (7)$$

That is, each producer in priority order receives its full bid quantity if residual demand permits, or the remaining residual if not, or zero once demand is satisfied. The last producer to be (partially) accepted exhausts the residual demand exactly.

Dispatch and Revenue

A producer i is dispatched for a quantity d_t^i defined as:

$$d_t^i = \begin{cases} q_t^i & \text{if } p_t^i < \lambda_t \quad (\text{infra-marginal, fully dispatched}) \\ d_t^{\pi_j} & \text{if } p_t^i = \lambda_t \quad (\text{marginal, priority-rationed via (7)}) \\ 0 & \text{if } p_t^i > \lambda_t \quad (\text{supra-marginal, not dispatched}) \end{cases} \quad (8)$$

All dispatched producers — whether fully or partially dispatched — are remunerated at the uniform clearing price λ_t . The wind producer's interval revenue is therefore:

$$R_t^W = \lambda_t \cdot d_t^W \quad (9)$$

and its total daily revenue remains:

$$R^W = \sum_{t=1}^{24} R_t^W \quad (10)$$

Implications for the Wind Producer

Under the price-taking assumption $p_t^W = 0$ (equation 14), the wind producer always bids at the price floor. Two cases arise:

- **If any conventional or solar producer also bids at zero**, the wind producer holds second priority behind solar, and may be partially dispatched or crowded out if solar alone satisfies residual demand.
- **In the typical case where conventional producers bid above zero**, the wind producer is either infra-marginal (fully dispatched at $q_t^W = \mu_q^W$) or, if total renewable supply at zero price exceeds demand, subject to priority rationing against the solar producer only. Given the priority $S \succ W$, solar is always satisfied before wind in such cases, exposing the wind producer to the greatest rationing risk within the zero-price cohort.

Given that actual wind-power production is stochastic, it may fall short of the quantity committed to dispatch one day in advance. If the wind-power producer under-delivers relative to its dispatched quantity, the shortfall incurs a constant penalty cost (\$/MWh). Please incorporate this penalty mechanism into your specification.

Then, please provide a complete model specification that specifies all key components, including the time scales, agents, state variables, control variables, processes, objectives, constants, uncertain parameters and stochastic variables.

Complete Agent-Based Model Specification: Wind-Power Producer in a Day-Ahead Spot Market

1. Time Scales

Scale	Description
Planning horizon	A single future day, indexed d
Trading intervals	$t \in \mathcal{T} = \{1, 2, \dots, 24\}$, each representing one clock hour
Bid submission	Once, before the day begins, in a single sealed-bid round covering all 24 intervals
Market clearing	Once per interval t , executed simultaneously for all t following bid submission
Physical delivery	Each interval t on the delivery day; actual wind production is realised here
Settlement	End of delivery day; revenues and penalty costs are computed against realised output

The separation between bid submission and physical delivery is the source of the wind producer's imbalance risk: bids are committed under uncertainty, while actual production is only revealed at delivery.

2. Agents

2.1 Producer Agents

Agent	Index	Technology	Behavioural Role
Conventional 1	C_1	Coal	Price-taking; bids at stochastic price, full capacity
Conventional 2	C_2	Coal	Price-taking; bids at stochastic price, full capacity
Conventional 3	C_3	Coal	Price-taking; bids at stochastic price, full capacity
Solar producer	S	Solar PV	Price-taking; bids deterministic cosine profile, stochastic price
Wind producer	W	Wind	Focal agent ; bids at zero price, mean capacity; subject to imbalance penalty

Each producer is an autonomous decision-maker that observes its own cost and resource parameters, forms beliefs about competitor behaviour, and independently submits a bid vector for the day.

2.2 Market Operator

A passive, non-strategic clearing agent that collects all bids, applies the merit-order algorithm with priority rationing, and returns the clearing price and dispatch quantities. It has no objective of its own and exercises no discretion beyond the clearing rules.

3. State Variables

State variables describe the condition of the system at each interval t before settlement is executed.

Symbol	Domain	Description
$\tilde{D}_t$	$\mathbb{R}_{>0}$	Aggregate electricity demand (MWh) in interval t
$\tilde{f}_t^W$	$\mathbb{R}_{\geq 0}$	Actual wind production (MWh) realised at delivery in interval t
λ_t	$\mathbb{R}_{\geq 0}$	Market-clearing price (\\$/MWh) in interval t ; an outcome of clearing
d_t^i	$[0, \bar{Q}^i]$	Dispatched quantity (MWh) of producer i in interval t ; an outcome of clearing
Δ_t^W	$\mathbb{R}_{\geq 0}$	Wind shortfall (MWh) in interval t ; $\Delta_t^W = \max(0, d_t^W - \tilde{f}_t^W)$

4. Control Variables

Control variables are the decision variables each producer agent chooses at bid submission, before any uncertainty is resolved.

For producer i and interval t :

Symbol	Domain	Description
q_t^i	$[0, \bar{Q}^i]$	Bid quantity (MWh)
p_t^i	$[\underline{p}, \bar{p}]$	Bid price (\\$/MWh); minimum acceptable remuneration

The wind producer's control vector for the day is:

$$\mathbf{u}^W = \{(q_t^W, p_t^W)\}_{t=1}^{24}$$

Under the price-taking assumption, both components are fixed constants rather than optimised variables (see Section 8):

$$p_t^W = 0, \quad q_t^W = \mu_q^W \quad \forall t \in \mathcal{T}$$

5. Processes

5.1 Bid Formation

Each producer i submits a sealed bid (q_t^i, p_t^i) for all $t \in \mathcal{T}$. Bids are submitted simultaneously; no producer observes competitors' bids before submission. Bid quantities and prices for each producer class are specified in Section 9.

5.2 Merit-Order Market Clearing

For each interval t , the market operator clears the market in two sequential stages.

Stage 1: Merit-Order Ranking and Price Setting

All bids are ranked in ascending order of bid price. Let σ_t denote this ranking, so that $p_t^{\sigma_t(1)} \leq p_t^{\sigma_t(2)} \leq \dots \leq p_t^{\sigma_t(5)}$. The operator cumulates quantities along the supply stack until aggregate supply meets demand. The **market-clearing price** is set at the bid price of the marginal producer:

$$\lambda_t = p_t^{\sigma_t(k^*)}, \quad k^* = \min \left\{ k : \sum_{j=1}^k q_t^{\sigma_t(j)} \geq \tilde{D}_t \right\} \quad (1)$$

Producers bidding strictly below λ_t are infra-marginal and fully dispatched. Producers bidding strictly above λ_t are not dispatched. The **marginal cohort** consists of all producers bidding exactly at λ_t :

$$\mathcal{M}_t = \{i : p_t^i = \lambda_t\} \quad (2)$$

Stage 2: Priority Dispatch Within the Marginal Cohort

When $|\mathcal{M}_t| > 1$ and the marginal cohort's collective capacity exceeds residual demand, producers are rationed by strict priority order. Residual demand at the margin is:

$$\tilde{D}_t^{\text{res}} = \tilde{D}_t - \sum_{\{i: p_t^i < \lambda_t\}} q_t^i \quad (3)$$

Producers in $\mathcal{M}_t$ are dispatched sequentially under the priority order $S \succ W \succ C_1 \succ C_2 \succ C_3$. Letting $\pi = (\pi_1, \pi_2, \dots)$ denote members of $\mathcal{M}_t$ sorted by priority, each receives:

$$d_t^{\pi_j} = \min \left(q_t^{\pi_j}, \max \left(0, \tilde{D}_t^{\text{res}} - \sum_{l < j} d_t^{\pi_l} \right) \right) \quad (4)$$

Full Dispatch Rule

Combining both stages, the dispatched quantity of producer i is:

$$d_t^i = \begin{cases} q_t^i & \text{if } p_t^i < \lambda_t \\ d_t^{\pi_j} & \text{if } p_t^i = \lambda_t \\ 0 & \text{if } p_t^i > \lambda_t \end{cases} \quad (5)$$

5.3 Physical Delivery and Shortfall Realisation

At delivery in interval t , each producer supplies electricity. For conventional and solar producers, delivery is assumed to equal the dispatched quantity. For the wind producer, actual production $\tilde{f}_t^W$ is stochastic and may fall short of the dispatched quantity d_t^W . The **wind shortfall** is:

$$\Delta_t^W = \max(0, d_t^W - \tilde{f}_t^W) \quad (6)$$

A shortfall of $\Delta_t^W > 0$ occurs whenever actual wind production is insufficient to cover the committed dispatch. There is no upside adjustment: if $\tilde{f}_t^W > d_t^W$, the surplus is curtailed and generates no additional revenue.

5.4 Settlement

All dispatched producers are remunerated at the uniform clearing price λ_t . The wind producer additionally incurs a penalty cost on any shortfall. Its **net revenue** in interval t is:

$$\Pi_t^W = \lambda_t \cdot d_t^W - \kappa \cdot \Delta_t^W \quad (7)$$

$$= \lambda_t \cdot d_t^W - \kappa \cdot \max(0, d_t^W - \tilde{f}_t^W) \quad (8)$$

where κ ($\$/\text{MWh}$) is the constant penalty rate. The first term is the gross dispatch revenue; the second is the imbalance penalty. Total **daily net revenue** is:

$$\Pi^W = \sum_{t=1}^{24} \Pi_t^W \quad (9)$$

6. Objective

At the single-day simulation stage, the model's purpose is **revenue estimation**: the wind producer's bid strategy is fixed by assumption (price-taking at zero, quantity equal to mean output), and the objective is to characterise the distribution of Π^W under uncertainty. The expected daily net revenue is:

$$\mathbb{E}[\Pi^W] = \sum_{t=1}^{24} \mathbb{E} \left[\lambda_t \cdot d_t^W - \kappa \cdot \max(0, d_t^W - \tilde{f}_t^W) \right] \quad (10)$$

The model can subsequently be extended to optimise $\mathbf{u}^W$ so as to maximise $\mathbb{E}[\Pi^W]$, subject to the feasibility constraints $0 \leq q_t^W \leq \mu_q^W$ and $p_t^W \in [\underline{p}, \bar{p}]$.

7. Constants and Fixed Parameters

Symbol	Description	Unit
$\bar{Q}^{C_k}$	Installed capacity of conventional producer $k \in \{C_1, C_2, C_3\}$	MWh
a	Vertical offset of solar generation profile	MWh
b	Amplitude of solar generation profile ($b > 0$)	MWh
κ	Penalty rate for wind shortfall	\$/MWh
$\underline{p}$	Market price floor	\$/MWh
$\bar{p}$	Market price cap	\$/MWh
$\$$	$\backslash\mathrm{mathcal{T}}$	\$
Priority order	$S \succ W \succ C_1 \succ C_2 \succ C_3$	—

8. Uncertain Parameters

Uncertain parameters are structural quantities that are not known with certainty but are fixed for a given simulation run. They are drawn from prior distributions and held constant across intervals t .

Symbol	Description	Unit
$\mu_p^{C_k}$	Mean bid price of conventional producer k	\$/MWh
σ_p^{2,C_k}	Variance of bid price of conventional producer k	(\$/MWh) ²
μ_p^S	Mean bid price of the solar producer	\$/MWh
$\sigma_p^{2,S}$	Variance of bid price of the solar producer	(\$/MWh) ²

9. Stochastic Variables

Stochastic variables are realised independently at each interval t and are the primary source of within-day randomness.

9.1 Demand

$$\tilde{D}_t \sim \mathcal{N}(\mu_D, \sigma_D^2) \quad \text{i.i.d.} \quad \forall t \in \mathcal{T} \quad (11)$$

Symbol	Description	Unit
μ_D	Mean aggregate demand	MWh
σ_D^2	Variance of aggregate demand	MWh ²

9.2 Conventional Producers

Bid quantities are deterministic; bid prices are stochastic:

$$q_t^{C_k} = \bar{Q}^{C_k} \quad (12)$$

$$\tilde{p}_t^{C_k} \sim \mathcal{N}(\mu_p^{C_k}, \sigma_p^{2,C_k}) \quad \text{i.i.d.} \quad \forall t \in \mathcal{T} \quad (13)$$

Realisations outside $[\underline{p}, \bar{p}]$ are truncated to the market price bounds.

9.3 Solar Producer

Bid quantities follow a deterministic diurnal cosine profile; bid prices are stochastic:

$$q_t^S = \max\left(0, a + b \cos\left(\frac{2\pi t}{24}\right)\right) \quad (14)$$

$$\tilde{p}_t^S \sim \mathcal{N}(\mu_p^S, \sigma_p^{2,S}) \text{ i.i.d. } \forall t \in \mathcal{T} \quad (15)$$

The floor at zero ensures no negative generation is bid. The zero-output window occurs when $a + b \cos(2\pi t/24) \leq 0$.

9.4 Wind Producer

Bid quantities are fixed at mean capacity; bid prices are set to zero:

$$q_t^W = \mu_q^W, \quad p_t^W = 0 \quad \forall t \in \mathcal{T} \quad (16)$$

Actual wind production is stochastic, realised at delivery:

$$\tilde{f}_t^W \sim \mathcal{N}(\mu_q^W, \sigma_q^{2,W}) \text{ i.i.d. } \forall t \in \mathcal{T} \quad (17)$$

Symbol	Description	Unit
μ_q^W	Mean wind production (= bid quantity)	MWh
$\sigma_q^{2,W}$	Variance of actual wind production	MWh ²

Realisations $\tilde{f}_t^W < 0$ are physically infeasible; provided σ_q^W / μ_q^W is small, their probability is negligible. A truncated normal $\mathcal{N}_{(0, \bar{Q}^W)}(\mu_q^W, \sigma_q^{2,W})$ may be substituted if tighter control is required.

9.5 Summary

Symbol	Agent	Type	Specification
$\tilde{D}_t$	Market	Stochastic	$\mathcal{N}(\mu_D, \sigma_D^2)$, i.i.d. across t
$q_t^{C_k}$	Conventional k	Constant	$\bar{Q}^{C_k}$
$\tilde{p}_t^{C_k}$	Conventional k	Stochastic	$\mathcal{N}(\mu_p^{C_k}, \sigma_p^{2,C_k})$, i.i.d. across t ; uncertain parameters
q_t^S	Solar	Deterministic	$\max(0, a + b \cos(2\pi t/24))$
$\tilde{p}_t^S$	Solar	Stochastic	$\mathcal{N}(\mu_p^S, \sigma_p^{2,S})$, i.i.d. across t ; uncertain parameters
q_t^W	Wind	Constant	μ_q^W
p_t^W	Wind	Constant	0 \$/MWh
$\tilde{f}_t^W$	Wind	Stochastic	$\mathcal{N}(\mu_q^W, \sigma_q^{2,W})$, i.i.d. across t ; realised at delivery

10. Key Modelling Assumptions

1. **Single-node network:** no transmission constraints; the market is copper-plate.
2. **Inelastic demand:** $\tilde{D}_t$ is exogenous and must be fully satisfied in every interval.
3. **Price-taking renewables:** wind and solar bid at or near zero marginal cost with no strategic mark-up.
4. **Price-taking conventionals:** conventional producers bid at a stochastic price drawn around their marginal cost; no strategic withholding of capacity.
5. **No inter-interval linkages:** each interval is cleared independently; no ramping, storage, or carry-over constraints.
6. **Asymmetric imbalance settlement:** only under-delivery by the wind producer is penalised; over-production is curtailed without compensation.
7. **No secondary or balancing market:** the penalty κ is a reduced-form representation of imbalance costs; no explicit balancing market is modelled.
8. **Independence across intervals:** all stochastic variables are drawn i.i.d. across t , abstracting away intra-day temporal correlations.

Wind-power producer's bid quantity and price should be treated as control variables, and thereby the objective should be framed as maximisation. Please refine your specification accordingly and present the complete specification.

Complete Agent-Based Model Specification: Wind-Power Producer in a Day-Ahead Spot Market

1. Time Scales

Scale	Description
Planning horizon	A single future day, indexed d
Trading intervals	$t \in \mathcal{T} = \{1, 2, \dots, 24\}$, each representing one clock hour
Bid submission	Once, before the day begins, in a single sealed-bid round covering all 24 intervals
Market clearing	Once per interval t , executed simultaneously for all t following bid submission
Physical delivery	Each interval t on the delivery day; actual wind production is realised here
Settlement	End of delivery day; revenues and penalty costs are computed against realised output

The separation between bid submission and physical delivery is the source of the wind producer's imbalance risk: bids are committed under uncertainty, while actual production is only revealed at delivery.

2. Agents

2.1 Producer Agents

Agent	Index	Technology	Behavioural Role
Conventional 1	C_1	Coal	Price-taking; bids at stochastic price, full capacity
Conventional 2	C_2	Coal	Price-taking; bids at stochastic price, full capacity
Conventional 3	C_3	Coal	Price-taking; bids at stochastic price, full capacity
Solar producer	S	Solar PV	Price-taking; bids deterministic cosine profile, stochastic price
Wind producer	W	Wind	Focal agent ; optimises bid quantity and price to maximise expected net revenue

Each producer is an autonomous decision-maker that observes its own cost and resource parameters, forms beliefs about competitor behaviour, and independently submits a bid vector for the day.

2.2 Market Operator

A passive, non-strategic clearing agent that collects all bids, applies the merit-order algorithm with priority rationing, and returns the clearing price and dispatch quantities. It has no objective of its own and exercises no discretion beyond the clearing rules.

3. State Variables

State variables describe the condition of the system at each interval t before settlement is executed.

Symbol	Domain	Description
$\tilde{D}_t$	$\mathbb{R}_{>0}$	Aggregate electricity demand (MWh) in interval t
$\tilde{f}_t^W$	$\mathbb{R}_{\geq 0}$	Actual wind production (MWh) realised at delivery in interval t
λ_t	$\mathbb{R}_{\geq 0}$	Market-clearing price (\$/MWh) in interval t ; an outcome of clearing
d_t^i	$[0, \bar{Q}^i]$	Dispatched quantity (MWh) of producer i in interval t ; an outcome of clearing
Δ_t^W	$\mathbb{R}_{\geq 0}$	Wind shortfall (MWh) in interval t ; $\Delta_t^W = \max(0, d_t^W - \tilde{f}_t^W)$

4. Control Variables

Control variables are the decision variables each producer agent chooses at bid submission, before any uncertainty is resolved.

For producer i and interval t :

Symbol	Domain	Description
q_t^i	$[0, \bar{Q}^i]$	Bid quantity (MWh)
p_t^i	$[\underline{p}, \bar{p}]$	Bid price (\$/MWh); minimum acceptable remuneration

For all competitor agents ($i \in \{C_1, C_2, C_3, S\}$), bids are governed by the fixed behavioural rules specified in Section 9. For the **wind producer**, both bid quantity and bid price are genuine optimisation variables. The wind producer's control vector for the day is:

$$\mathbf{u}^W = \{(q_t^W, p_t^W)\}_{t=1}^{24}$$

subject to the feasibility constraints:

$$0 \leq q_t^W \leq \mu_q^W \quad \forall t \in \mathcal{T} \quad (1)$$

$$\underline{p} \leq p_t^W \leq \bar{p} \quad \forall t \in \mathcal{T} \quad (2)$$

Constraint (1) bounds the bid quantity below by zero and above by the mean available wind capacity μ_q^W , reflecting the fact that the producer cannot reliably commit to more than its expected output. Constraint (2) confines the bid price to the permissible market range. The optimal values of $\mathbf{u}^W$ are determined by the maximisation problem stated in Section 6.

5. Processes

5.1 Bid Formation

Each producer i submits a sealed bid (q_t^i, p_t^i) for all $t \in \mathcal{T}$. Bids are submitted simultaneously; no producer observes competitors' bids before submission. Bid quantities and prices for each producer class are specified in Section 9. The wind producer selects its bid vector $\mathbf{u}^W$ so as to maximise the objective in Section 6.

5.2 Merit-Order Market Clearing

For each interval t , the market operator clears the market in two sequential stages.

Stage 1: Merit-Order Ranking and Price Setting

All bids are ranked in ascending order of bid price. Let σ_t denote this ranking, so that $p_t^{\sigma_t(1)} \leq p_t^{\sigma_t(2)} \leq \dots \leq p_t^{\sigma_t(5)}$. The operator cumulates quantities along the supply stack until aggregate supply meets demand. The **market-clearing price** is set at the bid price of the marginal producer:

$$\lambda_t = p_t^{\sigma_t(k^*)}, \quad k^* = \min \left\{ k : \sum_{j=1}^k q_t^{\sigma_t(j)} \geq \tilde{D}_t \right\} \quad (3)$$

Producers bidding strictly below λ_t are infra-marginal and fully dispatched. Producers bidding strictly above λ_t are not dispatched. The **marginal cohort** consists of all producers bidding exactly at λ_t :

$$\mathcal{M}_t = \{i : p_t^i = \lambda_t\} \quad (4)$$

Stage 2: Priority Dispatch Within the Marginal Cohort

When $|\mathcal{M}_t| > 1$ and the marginal cohort's collective capacity exceeds residual demand, producers are rationed by strict priority order. Residual demand at the margin is:

$$\tilde{D}_t^{\text{res}} = \tilde{D}_t - \sum_{\{i : p_t^i < \lambda_t\}} q_t^i \quad (5)$$

Producers in $\mathcal{M}_t$ are dispatched sequentially under the priority order $S \succ W \succ C_1 \succ C_2 \succ C_3$. Letting $\pi = (\pi_1, \pi_2, \dots)$ denote members of $\mathcal{M}_t$ sorted by priority, each receives:

$$d_t^{\pi_j} = \min \left(q_t^{\pi_j}, \max \left(0, \tilde{D}_t^{\text{res}} - \sum_{l < j} d_t^{\pi_l} \right) \right) \quad (6)$$

Full Dispatch Rule

Combining both stages, the dispatched quantity of producer i is:

$$d_t^i = \begin{cases} q_t^i & \text{if } p_t^i < \lambda_t \\ d_t^{\pi_j} & \text{if } p_t^i = \lambda_t \\ 0 & \text{if } p_t^i > \lambda_t \end{cases} \quad (7)$$

5.3 Physical Delivery and Shortfall Realisation

At delivery in interval t , each producer supplies electricity. For conventional and solar producers, delivery is assumed to equal the dispatched quantity. For the wind producer, actual production $\tilde{f}_t^W$ is stochastic and may fall short of the dispatched quantity d_t^W . The **wind shortfall** is:

$$\Delta_t^W = \max(0, d_t^W - \tilde{f}_t^W) \quad (8)$$

If $\tilde{f}_t^W > d_t^W$, the surplus is curtailed and generates no additional revenue.

5.4 Settlement

All dispatched producers are remunerated at the uniform clearing price λ_t . The wind producer additionally incurs a penalty cost on any shortfall. Its **net revenue** in interval t is:

$$\Pi_t^W = \lambda_t \cdot d_t^W - \kappa \cdot \Delta_t^W \quad (9)$$

$$= \lambda_t \cdot d_t^W - \kappa \cdot \max(0, d_t^W - \tilde{f}_t^W) \quad (10)$$

where κ ($\$/MWh$) is the constant penalty rate. The first term is the gross dispatch revenue; the second is the imbalance penalty. Total **daily net revenue** is:

$$\Pi^W = \sum_{t=1}^{24} \Pi_t^W \quad (11)$$

6. Objective

The wind producer selects its bid vector $\mathbf{u}^W = \{(q_t^W, p_t^W)\}_{t=1}^{24}$ to maximise expected daily net revenue:

$$\max_{\mathbf{u}^W} \mathbb{E}[\Pi^W] = \max_{\{q_t^W, p_t^W\}_{t=1}^{24}} \sum_{t=1}^{24} \mathbb{E} \left[\lambda_t \cdot d_t^W - \kappa \cdot \max \left(0, d_t^W - \tilde{f}_t^W \right) \right] \quad (12)$$

subject to the feasibility constraints (1) and (2). The expectation is taken jointly over all stochastic variables: demand $\tilde{D}_t$, actual wind production $\tilde{f}_t^W$, and competitor bid prices $\tilde{p}_t^{C_k}, \tilde{p}_t^S$.

The optimisation involves a fundamental trade-off governed by the bid variables:

- **Bid price p_t^W** : a higher bid price reduces the probability of dispatch (risk of earning zero revenue) but, when accepted, sets or raises the clearing price λ_t ; a lower bid price increases the probability of dispatch but may suppress λ_t if the wind producer becomes the marginal producer.
- **Bid quantity q_t^W** : a higher committed quantity increases potential revenue when dispatched but enlarges the expected shortfall $\mathbb{E}[\Delta_t^W]$ and thus the expected penalty cost; a lower committed quantity reduces imbalance risk but forgoes revenue on units that could have been reliably delivered.

Because λ_t and d_t^W are both non-linear functions of the full bid profile $\{(q_t^i, p_t^i)\}_i$ and the stochastic variables, a closed-form solution is generally intractable, and the expectation in (12) is evaluated by Monte Carlo simulation.

7. Constants and Fixed Parameters

Symbol	Description	Unit
$\bar{Q}^{C_k}$	Installed capacity of conventional producer $k \in \{C_1, C_2, C_3\}$	MWh
a	Vertical offset of solar generation profile	MWh
b	Amplitude of solar generation profile ($b > 0$)	MWh
κ	Penalty rate for wind shortfall	\$/MWh
$\underline{p}$	Market price floor	\$/MWh
$\bar{p}$	Market price cap	\$/MWh
μ_D	Mean aggregate demand	MWh
σ_D^2	Variance of aggregate demand	MWh ²
μ_q^W	Mean actual wind production (upper bound on bid quantity)	MWh
$\sigma_q^{2,W}$	Variance of actual wind production	MWh ²
\$	\mathcal{T}	\$
Priority order	$S \succ W \succ C_1 \succ C_2 \succ C_3$	—

8. Uncertain Parameters

Uncertain parameters are structural quantities that are not known with certainty but are fixed for a given simulation run. They are drawn from prior distributions before the day begins and held constant across all intervals t .

Symbol	Description	Unit
$\mu_p^{C_k}$	Mean bid price of conventional producer k	\$/MWh
σ_p^{2,C_k}	Variance of bid price of conventional producer k	(\$/MWh) ²
μ_p^S	Mean bid price of the solar producer	\$/MWh
$\sigma_p^{2,S}$	Variance of bid price of the solar producer	(\$/MWh) ²

9. Stochastic Variables

Stochastic variables are realised independently at each interval t and constitute the primary source of within-day randomness.

9.1 Demand

$$\tilde{D}_t \sim \mathcal{N}(\mu_D, \sigma_D^2) \quad \text{i.i.d.} \quad \forall t \in \mathcal{T} \quad (13)$$

Realisations $\tilde{D}_t \leq 0$ are physically infeasible; provided σ_D/μ_D is small, their probability is negligible. A truncated normal $\mathcal{N}_{(0,\infty)}(\mu_D, \sigma_D^2)$ may be substituted if required.

9.2 Conventional Producers

Bid quantities are deterministic and fixed at installed capacity; bid prices are stochastic:

$$q_t^{C_k} = \bar{Q}^{C_k} \quad (14)$$

$$\tilde{p}_t^{C_k} \sim \mathcal{N}(\mu_p^{C_k}, \sigma_p^{2,C_k}) \quad \text{i.i.d.} \quad \forall t \in \mathcal{T} \quad (15)$$

Realisations outside $[\underline{p}, \bar{p}]$ are truncated to the market price bounds.

9.3 Solar Producer

Bid quantities follow a deterministic diurnal cosine profile; bid prices are stochastic:

$$q_t^S = \max\left(0, a + b \cos\left(\frac{2\pi t}{24}\right)\right) \quad (16)$$

$$\tilde{p}_t^S \sim \mathcal{N}(\mu_p^S, \sigma_p^{2,S}) \quad \text{i.i.d.} \quad \forall t \in \mathcal{T} \quad (17)$$

The floor at zero ensures no negative generation is bid. The zero-output window occurs when $a + b \cos(2\pi t/24) \leq 0$.

9.4 Wind Producer

The wind producer's bid quantity and bid price are control variables, optimised per Section 6 subject to constraints (1) and (2). Actual wind production is stochastic, realised at delivery independently of the bid:

$$\tilde{f}_t^W \sim \mathcal{N}(\mu_q^W, \sigma_q^{2,W}) \quad \text{i.i.d.} \quad \forall t \in \mathcal{T} \quad (18)$$

The bid quantity $q_t^W \in [0, \mu_q^W]$ determines the producer's exposure to both dispatch revenue and imbalance penalties. Bidding $q_t^W = \mu_q^W$ maximises potential revenue but also maximises expected shortfall risk; bidding $q_t^W < \mu_q^W$ conservatively reduces imbalance risk at the cost of forgone revenue opportunities.

9.5 Summary

Symbol	Agent	Type	Specification
$\tilde{D}_t$	Market	Stochastic	$\mathcal{N}(\mu_D, \sigma_D^2)$, i.i.d. across t
$q_t^{C_k}$	Conventional k	Constant	$\bar{Q}^{C_k}$
$\tilde{p}_t^{C_k}$	Conventional k	Stochastic	$\mathcal{N}(\mu_p^{C_k}, \sigma_p^{2,C_k})$, i.i.d. across t ; uncertain parameters
q_t^S	Solar	Deterministic	$\max(0, a + b \cos(2\pi t/24))$
$\tilde{p}_t^S$	Solar	Stochastic	$\mathcal{N}(\mu_p^S, \sigma_p^{2,S})$, i.i.d. across t ; uncertain parameters
q_t^W	Wind	Control variable	$q_t^W \in [0, \mu_q^W]$; optimised per Section 6
p_t^W	Wind	Control variable	$p_t^W \in [\underline{p}, \bar{p}]$; optimised per Section 6
$\tilde{f}_t^W$	Wind	Stochastic	$\mathcal{N}(\mu_q^W, \sigma_q^{2,W})$, i.i.d. across t ; realised at delivery

10. Key Modelling Assumptions

1. **Single-node network:** no transmission constraints; the market is copper-plate.

2. **Inelastic demand:** $\bar{D}_t$ is exogenous and must be fully satisfied in every interval.
3. **Price-taking competitors:** conventional and solar producers bid without strategic mark-up; their bid prices reflect cost-based or rule-based behaviour as specified in Section 9.
4. **Strategic wind producer:** the wind producer optimises its bid quantity and price; it is the sole strategic agent in the model.
5. **No inter-interval linkages:** each interval is cleared independently; no ramping, storage, or carry-over constraints.
6. **Asymmetric imbalance settlement:** only under-delivery by the wind producer is penalised at the constant rate κ ; over-production relative to the dispatch commitment is curtailed without compensation.
7. **No secondary or balancing market:** the penalty κ is a reduced-form representation of imbalance costs; no explicit balancing market is modelled.
8. **Independence across intervals:** all stochastic variables are drawn i.i.d. across t , abstracting away intra-day temporal correlations.
9. **Intractability of closed-form solution:** because λ_t and d_t^W are non-linear, discontinuous functions of the stochastic variables and the full bid profile, the expectation in (12) is evaluated by Monte Carlo simulation.

Amplitude of solar generation profile b should be less than 0. Please refine your specification accordingly

The sign of b has a direct consequence for the physical interpretation of the solar generation profile, and the specification requires revision in two places: Section 7 (constants) and Section 9.3 (solar producer).

Revised Entry in Section 7 — Constants and Fixed Parameters

Symbol	Description	Unit
b	Amplitude of solar generation profile ($b < 0$)	MWh

Revised Section 9.3 — Solar Producer

Bid quantities follow a deterministic diurnal cosine profile; bid prices are stochastic:

$$q_t^S = \max\left(0, a + b \cos\left(\frac{2\pi t}{24}\right)\right) \quad (16)$$

$$\tilde{p}_t^S \sim \mathcal{N}(\mu_p^S, \sigma_p^{2,S}) \quad \text{i.i.d.} \quad \forall t \in \mathcal{T} \quad (17)$$

Interpretation of $b < 0$. Since $\cos(2\pi t/24)$ attains its maximum of $+1$ at $t = 0$ (midnight) and its minimum of -1 at $t = 12$ (noon), multiplying by $b < 0$ inverts the profile: the term $b \cos(2\pi t/24)$ is most negative at midnight and least negative at noon. The bid quantity q_t^S therefore reaches its peak at $t = 12$ and its trough at $t = 0$, which is consistent with the physical reality of solar irradiance peaking at solar noon and falling to zero overnight.

Zero-output window. The floor at zero ensures no negative generation is bid. The producer is inactive — $q_t^S = 0$ — during the overnight window where $a + b \cos(2\pi t/24) \leq 0$. For this window to be non-empty (i.e. for there to exist intervals with zero solar output, as physically required), the parameters must satisfy $a < |b|$, so that the profile dips to or below zero around midnight. If $a \geq |b|$, the profile remains strictly positive at all hours, which would be physically unrealistic and should be excluded by the choice of parameter values.

Various stakeholders approach this problem from different perspectives. Please revise your specification to use agents to represent two perspectives separately and independently: (1) the wind-power producer; and (2) the system regulator. In the revised specification, specify the common environment, including the agents, global constants, uncertain parameters and stochastic variables. For each perspective, specify the agent's key components, including state variables, control variables, processes and objectives.

Agent-Based Model Specification: Wind-Power Producer in a Day-Ahead Spot Market

A Dual-Perspective Formulation

PART I: COMMON ENVIRONMENT

I.1 Time Scales

Scale	Description
Planning horizon	A single future day, indexed d
Trading intervals	$t \in \mathcal{T} = \{1, 2, \dots, 24\}$, each representing one clock hour
Bid submission	Once, before the day begins, in a single sealed-bid round covering all 24 intervals
Market clearing	Once per interval t , executed simultaneously for all t following bid submission
Physical delivery	Each interval t on the delivery day; actual wind production is realised here
Settlement	End of delivery day; revenues, costs and penalties are computed against realised output

The separation between bid submission and physical delivery is the fundamental source of uncertainty in the model: all agents commit to decisions before stochastic variables are resolved.

I.2 Agents

Five producer agents and one market operator populate the shared environment. Two of the producer agents — the wind producer and the system regulator — are the focal agents of the dual-perspective formulation and are specified in full in Parts II and III respectively. The remaining agents are competitors or infrastructure, specified here.

Agent	Index	Type	Role
Wind producer	W	Focal — Perspective 1	Strategic bidder; maximises expected net revenue
System regulator	R	Focal — Perspective 2	Sets penalty rate; maximises social welfare
Conventional 1	C_1	Competitor	Price-taking; bids at stochastic price, full capacity
Conventional 2	C_2	Competitor	Price-taking; bids at stochastic price, full capacity
Conventional 3	C_3	Competitor	Price-taking; bids at stochastic price, full capacity
Solar producer	S	Competitor	Price-taking; bids deterministic cosine profile, stochastic price
Market operator	M	Infrastructure	Passive; applies merit-order clearing and priority rationing

I.3 Global Constants and Fixed Parameters

These parameters are common to both perspectives and invariant throughout the simulation.

Symbol	Description	Unit
$\bar{Q}^{C_k}$	Installed capacity of conventional producer $k \in \{C_1, C_2, C_3\}$	MWh
a	Vertical offset of solar generation profile	MWh
b	Amplitude of solar generation profile ($b < 0$)	MWh
$\underline{p}$	Market price floor	\$/MWh
$\bar{p}$	Market price cap	\$/MWh
μ_D	Mean aggregate demand, common to all intervals	MWh
σ_D^2	Variance of aggregate demand, common to all intervals	MWh ²
μ_q^W	Mean actual wind production; upper bound on wind bid quantity	MWh
$\sigma_q^{2,W}$	Variance of actual wind production	MWh ²
$\$$	$\backslash\mathrm{mathcal{T}}$	\$
Priority order	$S \succ W \succ C_1 \succ C_2 \succ C_3$	—

I.4 Uncertain Parameters

These structural quantities are not known with certainty but are fixed for a given simulation run. They are drawn from prior distributions before the day begins and held constant across all intervals t .

Symbol	Description	Unit
$\mu_p^{C_k}$	Mean bid price of conventional producer k	\$/MWh
σ_p^{2,C_k}	Variance of bid price of conventional producer k	(\$/MWh) ²
μ_p^S	Mean bid price of the solar producer	\$/MWh
$\sigma_p^{2,S}$	Variance of bid price of the solar producer	(\$/MWh) ²

I.5 Stochastic Variables

These variables are realised independently at each interval t and constitute the primary source of within-day randomness shared across both perspectives.

Demand

$$\tilde{D}_t \sim \mathcal{N}(\mu_D, \sigma_D^2) \quad \text{i.i.d.} \quad \forall t \in \mathcal{T} \quad (1)$$

Realisations $\tilde{D}_t \leq 0$ are physically infeasible; provided σ_D/μ_D is small, their probability is negligible. A truncated normal $\mathcal{N}_{(0,\infty)}(\mu_D, \sigma_D^2)$ may be substituted if required.

Conventional producers — bid quantities are deterministic; bid prices are stochastic:

$$q_t^{C_k} = \bar{Q}^{C_k} \quad (2)$$

$$\tilde{p}_t^{C_k} \sim \mathcal{N}(\mu_p^{C_k}, \sigma_p^{2,C_k}) \quad \text{i.i.d.} \quad \forall t \in \mathcal{T} \quad (3)$$

Realisations outside $[\underline{p}, \bar{p}]$ are truncated to the market price bounds.

Solar producer — bid quantities follow a deterministic diurnal cosine profile; bid prices are stochastic:

$$q_t^S = \max\left(0, a + b \cos\left(\frac{2\pi t}{24}\right)\right) \quad (4)$$

$$\tilde{p}_t^S \sim \mathcal{N}(\mu_p^S, \sigma_p^{2,S}) \text{ i.i.d. } \forall t \in \mathcal{T} \quad (5)$$

Since $b < 0$, the profile peaks at $t = 12$ (noon) and falls to zero during the overnight window where $a + b \cos(2\pi t/24) \leq 0$. The condition $a < |b|$ must hold for a non-trivial zero-output window to exist.

Wind producer — actual production is stochastic, realised at delivery:

$$\tilde{f}_t^W \sim \mathcal{N}(\mu_q^W, \sigma_q^{2,W}) \text{ i.i.d. } \forall t \in \mathcal{T} \quad (6)$$

Realisations $\tilde{f}_t^W < 0$ are physically infeasible; a truncated normal $\mathcal{N}_{(0, \bar{Q}^W)}(\mu_q^W, \sigma_q^{2,W})$ may be substituted if required.

Summary

Symbol	Agent	Specification
$\tilde{D}_t$	Market	$\mathcal{N}(\mu_D, \sigma_D^2)$, i.i.d. across t
$q_t^{C_k}$	Conventional k	Constant $\bar{Q}^{C_k}$
$\tilde{p}_t^{C_k}$	Conventional k	$\mathcal{N}(\mu_p^{C_k}, \sigma_p^{2,C_k})$, i.i.d. across t ; uncertain parameters
q_t^S	Solar	Deterministic $\max(0, a + b \cos(2\pi t/24))$, $b < 0$
$\tilde{p}_t^S$	Solar	$\mathcal{N}(\mu_p^S, \sigma_p^{2,S})$, i.i.d. across t ; uncertain parameters
$\tilde{f}_t^W$	Wind	$\mathcal{N}(\mu_q^W, \sigma_q^{2,W})$, i.i.d. across t ; realised at delivery

I.6 Market Clearing Process

The market operator clears each interval t in two sequential stages. This process is common to both perspectives and its outcomes — the clearing price λ_t and dispatch quantities d_t^i — are observed by all agents.

Stage 1: Merit-Order Ranking and Price Setting

Bids are ranked in ascending order of price. Let σ_t denote this ranking. The market-clearing price is:

$$\lambda_t = p_t^{\sigma_t(k^*)}, \quad k^* = \min \left\{ k : \sum_{j=1}^k q_t^{\sigma_t(j)} \geq \tilde{D}_t \right\} \quad (7)$$

The marginal cohort is:

$$\mathcal{M}_t = \{i : p_t^i = \lambda_t\} \quad (8)$$

Stage 2: Priority Dispatch Within the Marginal Cohort

Residual demand at the margin:

$$\tilde{D}_t^{\text{res}} = \tilde{D}_t - \sum_{\{i : p_t^i < \lambda_t\}} q_t^i \quad (9)$$

Producers in $\mathcal{M}_t$ are dispatched sequentially under $S \succ W \succ C_1 \succ C_2 \succ C_3$. Letting $\pi = (\pi_1, \pi_2, \dots)$ denote members of $\mathcal{M}_t$ sorted by priority:

$$d_t^{\pi_j} = \min \left(q_t^{\pi_j}, \max \left(0, \tilde{D}_t^{\text{res}} - \sum_{l < j} d_t^{\pi_l} \right) \right) \quad (10)$$

Full Dispatch Rule

$$d_t^i = \begin{cases} q_t^i & \text{if } p_t^i < \lambda_t \\ d_t^{\pi_j} & \text{if } p_t^i = \lambda_t \\ 0 & \text{if } p_t^i > \lambda_t \end{cases} \quad (11)$$

PART II: PERSPECTIVE 1 — WIND-POWER PRODUCER

The wind producer is a strategic, profit-maximising agent that selects its bid vector before uncertainty is resolved, subject to imbalance risk arising from the stochastic nature of wind generation.

II.1 State Variables

Symbol	Domain	Description
$\tilde{f}_t^W$	$\mathbb{R}_{\geq 0}$	Actual wind production (MWh) realised at delivery in interval t
λ_t	$\mathbb{R}_{\geq 0}$	Market-clearing price (\\$/MWh) in interval t ; observed post-clearing
d_t^W	$[0, \mu_q^W]$	Dispatched quantity (MWh) of the wind producer in interval t ; outcome of clearing
Δ_t^W	$\mathbb{R}_{\geq 0}$	Wind shortfall (MWh); $\Delta_t^W = \max(0, d_t^W - \tilde{f}_t^W)$

II.2 Control Variables

The wind producer's bid vector $\mathbf{u}^W = \{(q_t^W, p_t^W)\}_{t=1}^{24}$ is chosen before any uncertainty is resolved.

Symbol	Domain	Description
q_t^W	$[0, \mu_q^W]$	Bid quantity (MWh) in interval t
p_t^W	$[\underline{p}, \bar{p}]$	Bid price (\\$/MWh) in interval t

Feasibility constraints:

$$0 \leq q_t^W \leq \mu_q^W \quad \forall t \in \mathcal{T} \quad (12)$$

$$\underline{p} \leq p_t^W \leq \bar{p} \quad \forall t \in \mathcal{T} \quad (13)$$

Constraint (12) ensures the producer does not commit to more than its mean available capacity. Constraint (13) confines the bid price to the permissible market range.

II.3 Processes

Bid submission. The wind producer submits (q_t^W, p_t^W) for all $t \in \mathcal{T}$ simultaneously with all other producers, prior to market clearing.

Physical delivery and shortfall realisation. At delivery in interval t , actual production $\tilde{f}_t^W$ is realised. The shortfall is:

$$\Delta_t^W = \max(0, d_t^W - \tilde{f}_t^W) \quad (14)$$

Surplus production above d_t^W is curtailed and generates no additional revenue.

Settlement. The wind producer's net revenue in interval t is:

$$\Pi_t^W = \lambda_t \cdot d_t^W - \kappa \cdot \Delta_t^W \quad (15)$$

$$= \lambda_t \cdot d_t^W - \kappa \cdot \max(0, d_t^W - \tilde{f}_t^W) \quad (16)$$

where κ (\\$/MWh) is the constant penalty rate set by the regulator. Total daily net revenue is:

$$\Pi^W = \sum_{t=1}^{24} \Pi_t^W \quad (17)$$

II.4 Objective

The wind producer selects its bid vector $\mathbf{u}^W$ to maximise expected daily net revenue:

$$\max_{\{q_t^W, p_t^W\}_{t=1}^{24}} \mathbb{E}[\Pi^W] = \max_{\{q_t^W, p_t^W\}_{t=1}^{24}} \sum_{t=1}^{24} \mathbb{E} \left[\lambda_t \cdot d_t^W - \kappa \cdot \max(0, d_t^W - \tilde{f}_t^W) \right] \quad (18)$$

subject to constraints (12) and (13). The expectation is taken jointly over demand $\tilde{D}_t$, actual wind production $\tilde{f}_t^W$, and competitor bid prices $\tilde{p}_t^{Ck}, \tilde{p}_t^S$.

The optimisation involves a fundamental trade-off on each control variable:

- **Bid price** p_t^W : a higher bid price reduces the probability of dispatch but raises the clearing price λ_t when accepted; a lower bid price increases dispatch probability but may depress λ_t if the wind producer becomes the marginal producer.
- **Bid quantity** q_t^W : a higher committed quantity increases potential dispatch revenue but enlarges expected shortfall $\mathbb{E}[\Delta_t^W]$ and thereby expected penalty costs; a lower committed quantity conservatively reduces imbalance risk at the cost of forgone revenue.

Because λ_t and d_t^W are non-linear, discontinuous functions of the stochastic variables and the full bid profile, the expectation in (18) is generally intractable in closed form and is evaluated by Monte Carlo simulation.

PART III: PERSPECTIVE 2 — SYSTEM REGULATOR

The system regulator is a welfare-maximising agent that sets the penalty rate κ applied to wind shortfalls. The regulator does not participate in bidding but observes market outcomes and seeks to align the wind producer's private incentives with broader system objectives.

III.1 State Variables

Symbol	Domain	Description
λ_t	$\mathbb{R}_{\geq 0}$	Market-clearing price (\\$/MWh) in interval t ; observed post-clearing
d_t^W	$[0, \mu_q^W]$	Dispatched quantity (MWh) of the wind producer; outcome of clearing
$\tilde{f}_t^W$	$\mathbb{R}_{\geq 0}$	Actual wind production (MWh) realised at delivery in interval t
Δ_t^W	$\mathbb{R}_{\geq 0}$	Wind shortfall (MWh); $\Delta_t^W = \max(0, d_t^W - \tilde{f}_t^W)$
C_t^{sys}	$\mathbb{R}_{\geq 0}$	System balancing cost (MWh) incurred to cover wind shortfall in interval t

III.2 Control Variables

The regulator's sole control variable is the penalty rate imposed on wind shortfalls:

Symbol	Domain	Description
κ	$[\underline{\kappa}, \bar{\kappa}]$	Penalty rate (\\$/MWh) applied to wind shortfall Δ_t^W

Feasibility constraint:

$$\underline{\kappa} \leq \kappa \leq \bar{\kappa} \quad (19)$$

where $\underline{\kappa} \geq 0$ is the minimum admissible penalty (a floor of zero permits no penalty) and $\bar{\kappa}$ is a regulatory ceiling. Unlike the wind producer's control vector, κ is a scalar set once for the day and applied uniformly across all intervals $t \in \mathcal{T}$.

III.3 Processes

Penalty design. Before the day begins, the regulator announces κ , which is observed by all producers prior to bid submission. The penalty rate thus enters the wind producer's objective (18) as a known parameter and influences its optimal bid vector $\mathbf{u}^W(\kappa)$.

Shortfall cost realisation. When the wind producer under-delivers in interval t , the system must procure replacement energy at the spot balancing cost c_t^{bal} (\\$/MWh) to maintain supply-demand balance. The system balancing cost is:

$$C_t^{\text{sys}} = c_t^{\text{bal}} \cdot \Delta_t^W \quad (20)$$

The penalty revenue collected from the wind producer is $\kappa \cdot \Delta_t^W$, which may partially or fully offset C_t^{sys} depending on the relationship between κ and c_t^{bal} .

Welfare accounting. The regulator evaluates social welfare as the sum of producer surplus and consumer surplus, net of system balancing costs. Total daily social welfare is:

$$\mathcal{W} = \underbrace{\sum_{t=1}^{24} \sum_i \Pi_t^i}_{\text{aggregate producer surplus}} + \underbrace{\sum_{t=1}^{24} \left(V \cdot \tilde{D}_t - \lambda_t \cdot \tilde{D}_t \right)}_{\text{consumer surplus}} - \underbrace{\sum_{t=1}^{24} C_t^{\text{sys}}}_{\text{balancing costs}} \quad (21)$$

where V (\\$/MWh) is the consumers' constant marginal value of electricity. The penalty payments $\kappa \cdot \Delta_t^W$ are transfers between the wind producer and the system and therefore net out of the social welfare calculation.

III.4 Objective

The regulator selects the penalty rate κ to maximise expected social welfare, internalising the effect of κ on the wind producer's bidding behaviour $\mathbf{u}^W(\kappa)$:

$$\max_{\kappa \in [\underline{\kappa}, \bar{\kappa}]} \mathbb{E}[\mathcal{W} \mid \mathbf{u}^W(\kappa)] \quad (22)$$

subject to constraint (19). The expectation is taken jointly over all stochastic variables in the common environment.

The regulator's problem is a **Stackelberg leadership problem**: the regulator moves first by announcing κ ; the wind producer then responds optimally with $\mathbf{u}^W(\kappa)$. The regulator must therefore anticipate the wind producer's best response when selecting κ . The key trade-off is:

- **Too low a penalty** ($\kappa \ll c_t^{\text{bal}}$): the wind producer under-prices imbalance risk, bids aggressively, and generates frequent large shortfalls, imposing high balancing costs on the system.
 - **Too high a penalty** ($\kappa \gg c_t^{\text{bal}}$): the wind producer bids conservatively, reducing shortfall risk but suppressing renewable dispatch and lowering consumer surplus and wind producer revenue.
 - **Optimal penalty** (κ^*): aligns the wind producer's private marginal cost of imbalance with the system's marginal balancing cost, internalising the externality.
-

PART IV: KEY MODELLING ASSUMPTIONS

1. **Single-node network**: no transmission constraints; the market is copper-plate.
2. **Inelastic demand**: $\tilde{D}_t$ is exogenous and must be fully satisfied in every interval.
3. **Price-taking competitors**: conventional and solar producers bid without strategic mark-up; their behaviour follows the fixed rules in Section I.5.

4. **Strategic wind producer:** the wind producer is the sole strategic bidding agent; it optimises $\mathbf{u}^W$ given κ .
5. **Stackelberg structure:** the regulator is the leader; the wind producer is the follower. The regulator anticipates the wind producer's best response.
6. **No inter-interval linkages:** each interval is cleared independently; no ramping, storage, or carry-over constraints.
7. **Asymmetric imbalance settlement:** only under-delivery by the wind producer is penalised; over-production is curtailed without compensation.
8. **Penalty as a transfer:** the penalty payment $\kappa \cdot \Delta_t^W$ is a transfer from wind producer to system and is welfare-neutral in aggregate; system balancing costs C_t^{sys} represent the true social cost of shortfalls.
9. **Independence across intervals:** all stochastic variables are drawn i.i.d. across t , abstracting away intra-day temporal correlations.
10. **Monte Carlo evaluation:** because λ_t and d_t^W are non-linear, discontinuous functions of the stochastic variables and the full bid profile, expectations in both (18) and (22) are evaluated by Monte Carlo simulation.

To ensure internal consistency across these perspectives, please compose them into a unified agent-based model. The composed model should: 1. embed each perspective's components, including state variables, control variables, processes and objectives; and 2. enable modular evaluation of each perspective, where other perspectives' controls are treated as exogenous inputs. Please specify the key components of the final model according to the ODD protocol.

Unified Agent-Based Model: Wind-Power Producer in a Day-Ahead Spot Market

Specification According to the ODD Protocol

SECTION 1: OVERVIEW

1.1 Purpose

The model simulates a single day of operations in a day-ahead electricity spot market comprising five competing producers. Its primary purpose is to support the economic assessment of a wind-power investment by jointly characterising two interdependent perspectives:

- **Perspective W** (*Wind-power producer*): determine the bid vector — quantity and price for each hourly interval — that maximises the wind producer's expected daily net revenue, subject to imbalance penalties arising from stochastic wind production.
- **Perspective R** (*System regulator*): determine the penalty rate applied to wind shortfalls that maximises expected social welfare, anticipating the wind producer's strategic bidding response.

The two perspectives are embedded within a single, internally consistent model. The composed model admits modular evaluation: either perspective's control variable may be fixed as an exogenous input while the other is optimised, or both may be co-optimised within a Stackelberg equilibrium framework. This supports stepwise analysis, sensitivity testing, and investment scenario assessment.

1.2 Entities, State Variables, and Scales

1.2.1 Time Scales

Scale	Description
Planning horizon	A single future day, indexed d

Scale	Description
Trading intervals	$t \in \mathcal{T} = \{1, 2, \dots, 24\}$, each representing one clock hour
Bid submission	Once, before the day begins, in a single sealed-bid round covering all 24 intervals
Market clearing	Once per interval t , executed simultaneously for all t following bid submission
Physical delivery	Each interval t on the delivery day; actual wind production is realised here
Settlement	End of delivery day; revenues, costs and penalties are computed against realised output

1.2.2 Entities and State Variables

The model contains seven entities: five producer agents, one market operator, and one system regulator. Each entity's state is described by the variables below.

Wind Producer (W) — Perspective W focal agent

Symbol	Domain	Description
$\tilde{f}_t^W$	$\mathbb{R}_{\geq 0}$	Actual wind production (MWh) realised at delivery in interval t
λ_t	$\mathbb{R}_{\geq 0}$	Market-clearing price (\\$/MWh) in interval t ; observed post-clearing
d_t^W	$[0, \mu_q^W]$	Dispatched quantity (MWh); outcome of market clearing
Δ_t^W	$\mathbb{R}_{\geq 0}$	Wind shortfall (MWh); $\Delta_t^W = \max(0, d_t^W - \tilde{f}_t^W)$
Π_t^W	$\mathbb{R}$	Net revenue (\\$) in interval t ; outcome of settlement
Π^W	$\mathbb{R}$	Total daily net revenue (\\$); aggregated across all t

System Regulator (R) — Perspective R focal agent

Symbol	Domain	Description
λ_t	$\mathbb{R}_{\geq 0}$	Market-clearing price (\\$/MWh) in interval t ; observed post-clearing
d_t^W	$[0, \mu_q^W]$	Dispatched wind quantity (MWh); observed post-clearing
$\tilde{f}_t^W$	$\mathbb{R}_{\geq 0}$	Actual wind production (MWh); observed post-delivery
Δ_t^W	$\mathbb{R}_{\geq 0}$	Wind shortfall (MWh); $\Delta_t^W = \max(0, d_t^W - \tilde{f}_t^W)$
C_t^{sys}	$\mathbb{R}_{\geq 0}$	System balancing cost (\\$) in interval t ; $C_t^{\text{sys}} = c_t^{\text{bal}} \cdot \Delta_t^W$
$\mathcal{W}$	$\mathbb{R}$	Total daily social welfare (\\$); aggregated across all t

Conventional Producers ($C_k, k \in \{1, 2, 3\}$)

Symbol	Domain	Description
$q_t^{C_k}$	$\{\bar{Q}^{C_k}\}$	Bid quantity (MWh); fixed at installed capacity
$\tilde{p}_t^{C_k}$	$[\underline{p}, \bar{p}]$	Bid price (\\$/MWh); stochastic, drawn each interval
$d_t^{C_k}$	$[0, \bar{Q}^{C_k}]$	Dispatched quantity (MWh); outcome of clearing

Solar Producer (S)

Symbol	Domain	Description
q_t^S	$\mathbb{R}_{\geq 0}$	Bid quantity (MWh); deterministic cosine profile
$\tilde{p}_t^S$	$[\underline{p}, \bar{p}]$	Bid price (\\$/MWh); stochastic, drawn each interval
d_t^S	$[0, q_t^S]$	Dispatched quantity (MWh); outcome of clearing

Market Operator (M)

Symbol	Domain	Description
λ_t	$\mathbb{R}_{\geq 0}$	Market-clearing price (\$/MWh); computed each interval
$\mathcal{M}_t$	$\subseteq \{W, S, C_1, C_2, C_3\}$	Marginal cohort in interval t
$\tilde{D}_t^{\text{res}}$	$\mathbb{R}_{\geq 0}$	Residual demand (MWh) at the margin in interval t

1.3 Process Overview and Scheduling

Processes are executed in the following strict sequential order within each simulated day. There is no iteration within a day; the sequence runs once.

Step	Stage	Executing Entity	Description
1	Penalty announcement	Regulator R	Sets and announces κ before any bids are submitted
2	Stochastic draw	Environment	Uncertain parameters $\mu_p^{C_k}, \sigma_p^{2,C_k}, \mu_p^S, \sigma_p^{2,S}$ drawn from priors
3	Bid formation	All producers	Each producer submits (q_t^i, p_t^i) for all $t \in \mathcal{T}$; sealed, simultaneous
4	Market clearing	Market operator M	Merit-order clearing and priority rationing for each t ; λ_t and d_t^i determined
5	Physical delivery	All producers	Producers supply electricity; $\tilde{f}_t^W$ realised for wind; shortfall Δ_t^W computed
6	Settlement	Market operator M	Revenue Π_t^W computed for all producers; balancing cost C_t^{sys} computed
7	Welfare accounting	Regulator R	Social welfare $\mathcal{W}$ computed from market outcomes and balancing costs

Step 1 precedes Step 3 to preserve the Stackelberg structure: κ is a public commitment that the wind producer treats as known when optimising $\mathbf{u}^W$.

SECTION 2: DESIGN CONCEPTS

2.1 Basic Principles

The model is grounded in three core principles. First, **merit-order market clearing**: the market operator dispatches producers in ascending order of bid price until demand is satisfied, setting a uniform clearing price at the marginal bid. This is a standard representation of competitive day-ahead electricity markets. Second, **strategic optimisation under uncertainty**: the wind producer maximises expected net revenue by choosing bid quantity and price before stochastic variables are resolved, accepting a trade-off between dispatch probability and imbalance risk. Third, **Stackelberg regulation**: the regulator acts as a first-mover leader, setting the penalty rate κ in anticipation of the wind producer's best-response bidding strategy, thereby internalising the social cost of wind shortfalls.

2.2 Emergence

The market-clearing price λ_t and dispatch quantities d_t^i are emergent outcomes: they are not prescribed but arise from the interaction of all producers' bids and the stochastic demand realisation through the merit-order mechanism. The wind producer's expected net revenue and the system's expected social welfare are likewise emergent, arising from the joint distribution of clearing outcomes across intervals and simulation runs.

2.3 Adaptation

The wind producer adapts its bid vector $\mathbf{u}^W = \{(q_t^W, p_t^W)\}_{t=1}^{24}$ in response to the announced penalty rate κ . As κ increases, the wind producer's optimal response is to reduce bid quantity q_t^W (to limit imbalance exposure) and/or adjust bid price p_t^W (to influence dispatch probability). This adaptive best-response mapping $\mathbf{u}^W(\kappa)$ is the behavioural channel through which the regulator's instrument affects market outcomes. Competitor agents do not adapt; their behaviour follows fixed stochastic rules.

2.4 Objectives

Each focal agent has a distinct, formally specified objective:

- **Wind producer:** maximise expected daily net revenue $\mathbb{E}[\Pi^W]$ by choosing $\mathbf{u}^W$, given κ and beliefs about competitor behaviour.
- **System regulator:** maximise expected daily social welfare $\mathbb{E}[\mathcal{W}]$ by choosing κ , anticipating $\mathbf{u}^W(\kappa)$.

Non-focal competitor agents (conventional and solar producers) do not optimise; they follow the behavioural rules specified in Section 3.4.

2.5 Learning

No within-day or across-day learning is implemented in the single-day model. The wind producer's optimal bid vector is determined analytically or via Monte Carlo optimisation prior to the simulated day, treating competitor behaviour and stochastic parameters as fixed. Extensions to multi-day scenarios may incorporate learning via reinforcement learning or Bayesian updating of beliefs about competitor bid distributions.

2.6 Prediction

The wind producer forms implicit predictions about competitor bid prices and demand realisations when solving its optimisation problem. These predictions take the form of distributional beliefs consistent with the stochastic specifications in Section 3.5, rather than point forecasts. The regulator similarly forms distributional predictions about all stochastic variables when evaluating expected social welfare.

2.7 Sensing

- The **wind producer** observes: (i) the announced penalty rate κ before bidding; (ii) its own mean capacity μ_q^W and variance $\sigma_q^{2,W}$; (iii) the distributional parameters of competitor bids (as beliefs, not direct observations); and (iv) the clearing price λ_t and its own dispatch d_t^W post-clearing, and actual production $\tilde{f}_t^W$ at delivery.
- The **system regulator** observes: (i) the wind producer's best-response function $\mathbf{u}^W(\kappa)$; and (ii) all post-clearing and post-delivery outcomes $\lambda_t, d_t^W, \tilde{f}_t^W, \Delta_t^W, C_t^{\text{sys}}$.
- **Competitor agents** do not observe each other's bids; all bids are sealed at submission.

2.8 Interaction

Agents interact indirectly through the market mechanism. No bilateral or direct agent-to-agent communication occurs. The market operator aggregates all bids and returns clearing outcomes to all agents simultaneously. The regulator interacts with the wind producer only through the public announcement of κ in Step 1.

2.9 Stochasticity

The model contains four sources of stochasticity, all specified as normal distributions:

Source	Variable	Distribution
Aggregate demand	$\tilde{D}_t$	$\mathcal{N}(\mu_D, \sigma_D^2)$, i.i.d. across t
Conventional bid prices	$\tilde{p}_t^{C_k}$	$\mathcal{N}(\mu_p^{C_k}, \sigma_p^{2,C_k})$, i.i.d. across t ; uncertain parameters
Solar bid price	$\tilde{p}_t^S$	$\mathcal{N}(\mu_p^S, \sigma_p^{2,S})$, i.i.d. across t ; uncertain parameters
Wind production	$\tilde{f}_t^W$	$\mathcal{N}(\mu_q^W, \sigma_q^{2,W})$, i.i.d. across t

Uncertain parameters $\mu_p^{C_k}, \sigma_p^{2,C_k}, \mu_p^S, \sigma_p^{2,S}$ are drawn once per simulation run from their respective priors before any interval-level stochastic variables are realised.

2.10 Collectives

No coalitions or collective entities are formed. All producers act independently.

2.11 Observation

The following quantities are recorded at the end of each simulated day and used to evaluate the objectives of both perspectives:

Observable	Perspective	Use
λ_t, d_t^W for all t	W, R	Revenue and welfare computation
$\tilde{f}_t^W, \Delta_t^W$ for all t	W, R	Shortfall and penalty computation
Π_t^W, Π^W	W	Wind producer objective evaluation
$C_t^{\text{sys}}, \mathcal{W}$	R	Regulator objective evaluation
$\mathbb{E}[\Pi^W], \mathbb{E}[\mathcal{W}]$	W, R	Monte Carlo estimates of expected objectives

SECTION 3: DETAILS

3.1 Initialisation

Before the simulated day begins, the following quantities are initialised:

- Regulator sets κ :** either optimised per the regulator's objective (Perspective R active) or fixed as an exogenous input (Perspective R passive).
- Wind producer sets $\mathbf{u}^W$:** either optimised per the wind producer's objective (Perspective W active) or fixed as an exogenous input (Perspective W passive).
- Uncertain parameters are drawn:** $\mu_p^{C_k} \sim \pi(\mu_p^{C_k}), \sigma_p^{2,C_k} \sim \pi(\sigma_p^{2,C_k}), \mu_p^S \sim \pi(\mu_p^S), \sigma_p^{2,S} \sim \pi(\sigma_p^{2,S})$ from their respective prior distributions; held fixed for the remainder of the run.
- Interval-level stochastic variables are not yet drawn:** $\tilde{D}_t, \tilde{p}_t^{C_k}, \tilde{p}_t^S, \tilde{f}_t^W$ are drawn at the point of use within Steps 4 and 5.

Modular evaluation modes:

Mode	κ	$\mathbf{u}^W$	Purpose
Perspective W only	Exogenous (fixed)	Optimised	Wind producer's optimal bidding given a prescribed penalty
Perspective R only	Optimised	Exogenous (fixed)	Regulator's optimal penalty given a prescribed bid strategy
Composed (Stackelberg)	Optimised	Optimised via best response	Joint equilibrium; regulator leads, wind producer follows
Simulation only	Exogenous (fixed)	Exogenous (fixed)	Revenue and welfare estimation for fixed scenario inputs

3.2 Global Constants and Fixed Parameters

Symbol	Description	Unit
$\bar{Q}^{C_k}$	Installed capacity of conventional producer k	MWh
a	Vertical offset of solar generation profile	MWh
b	Amplitude of solar generation profile ($b < 0$)	MWh
$\underline{p}$	Market price floor	\$/MWh
$\bar{p}$	Market price cap	\$/MWh
$\underline{\kappa}$	Minimum admissible penalty rate	\$/MWh
$\bar{\kappa}$	Maximum admissible penalty rate	\$/MWh

Symbol	Description	Unit
c_t^{bal}	System balancing cost in interval t	\$/MWh
V	Consumers' constant marginal value of electricity	\$/MWh
μ_D	Mean aggregate demand	MWh
σ_D^2	Variance of aggregate demand	MWh ²
μ_q^W	Mean actual wind production; upper bound on bid quantity	MWh
$\sigma_q^{2,W}$	Variance of actual wind production	MWh ²
$\$$	$\backslash\text{mathcal{T}$	\$
Priority order	$S \succ W \succ C_1 \succ C_2 \succ C_3$	—

3.3 Uncertain Parameters

Drawn once per simulation run from prior distributions; held constant across all intervals t .

Symbol	Description	Unit
$\mu_p^{C_k}$	Mean bid price of conventional producer k	\$/MWh
σ_p^{2,C_k}	Variance of bid price of conventional producer k	(\$/MWh) ²
μ_p^S	Mean bid price of the solar producer	\$/MWh
$\sigma_p^{2,S}$	Variance of bid price of the solar producer	(\$/MWh) ²

3.4 Submodel A: Bid Formation

Conventional producers. For each $k \in \{C_1, C_2, C_3\}$ and each $t \in \mathcal{T}$:

$$q_t^{C_k} = \bar{Q}^{C_k} \quad (1)$$

$$\tilde{p}_t^{C_k} \sim \mathcal{N}(\mu_p^{C_k}, \sigma_p^{2,C_k}), \quad \text{truncated to } [\underline{p}, \bar{p}] \quad (2)$$

Solar producer. For each $t \in \mathcal{T}$:

$$q_t^S = \max\left(0, a + b \cos\left(\frac{2\pi t}{24}\right)\right), \quad b < 0 \quad (3)$$

$$\tilde{p}_t^S \sim \mathcal{N}(\mu_p^S, \sigma_p^{2,S}), \quad \text{truncated to } [\underline{p}, \bar{p}] \quad (4)$$

Since $b < 0$, the cosine profile peaks at $t = 12$ (noon) and is zero in the overnight window where $a + b \cos(2\pi t/24) \leq 0$; the condition $a < |b|$ must hold for this window to be non-empty.

Wind producer. For each $t \in \mathcal{T}$, the bid vector $\mathbf{u}^W = \{(q_t^W, p_t^W)\}_{t=1}^{24}$ is either optimised (Perspective W active) or fixed as an exogenous input (Perspective W passive), subject to:

$$0 \leq q_t^W \leq \mu_q^W \quad (5)$$

$$\underline{p} \leq p_t^W \leq \bar{p} \quad (6)$$

3.5 Submodel B: Market Clearing

Executed by the market operator for each interval $t \in \mathcal{T}$.

Demand realisation:

$$\tilde{D}_t \sim \mathcal{N}(\mu_D, \sigma_D^2) \quad (7)$$

Stage 1 — Merit-order ranking and price setting. Rank all bids in ascending order of price under permutation σ_t . Set the market-clearing price:

$$\lambda_t = p_t^{\sigma_t(k^*)}, \quad k^* = \min \left\{ k : \sum_{j=1}^k q_t^{\sigma_t(j)} \geq \tilde{D}_t \right\} \quad (8)$$

Identify the marginal cohort:

$$\mathcal{M}_t = \{i : p_t^i = \lambda_t\} \quad (9)$$

Stage 2 — Priority rationing within the marginal cohort. Compute residual demand at the margin:

$$\tilde{D}_t^{\text{res}} = \tilde{D}_t - \sum_{\{i : p_t^i < \lambda_t\}} q_t^i \quad (10)$$

Dispatch members of $\mathcal{M}_t$ sequentially under $S \succ W \succ C_1 \succ C_2 \succ C_3$. Let $\pi = (\pi_1, \pi_2, \dots)$ denote the priority-ordered members of $\mathcal{M}_t$:

$$d_t^{\pi_j} = \min \left(q_t^{\pi_j}, \max \left(0, \tilde{D}_t^{\text{res}} - \sum_{l < j} d_t^{\pi_l} \right) \right) \quad (11)$$

Full dispatch rule:

$$d_t^i = \begin{cases} q_t^i & \text{if } p_t^i < \lambda_t \\ d_t^{\pi_j} & \text{if } p_t^i = \lambda_t \\ 0 & \text{if } p_t^i > \lambda_t \end{cases} \quad (12)$$

3.6 Submodel C: Physical Delivery and Settlement

Wind production realisation. At delivery in interval t :

$$\tilde{f}_t^W \sim \mathcal{N}(\mu_q^W, \sigma_q^{2,W}) \quad (13)$$

Wind shortfall:

$$\Delta_t^W = \max(0, d_t^W - \tilde{f}_t^W) \quad (14)$$

Surplus production above d_t^W is curtailed without compensation.

Wind producer net revenue (Perspective W):

$$\Pi_t^W = \lambda_t \cdot d_t^W - \kappa \cdot \Delta_t^W \quad (15)$$

$$\Pi^W = \sum_{t=1}^{24} \Pi_t^W \quad (16)$$

System balancing cost (Perspective R):

$$C_t^{\text{sys}} = c_t^{\text{bal}} \cdot \Delta_t^W \quad (17)$$

Social welfare (Perspective R):

$$\mathcal{W} = \sum_{t=1}^{24} \left[V \cdot \tilde{D}_t - \lambda_t \cdot \tilde{D}_t + \sum_i \Pi_t^i - C_t^{\text{sys}} \right] \quad (18)$$

The penalty payments $\kappa \cdot \Delta_t^W$ are transfers between the wind producer and the system and therefore net out of $\mathcal{W}$; only the true balancing cost C_t^{sys} enters as a social loss.

3.7 Submodel D: Perspective-Specific Optimisation

Perspective W — Wind Producer Optimisation

Given announced κ and distributional beliefs over all stochastic variables, the wind producer solves:

$$\max_{\{q_t^W, p_t^W\}_{t=1}^{24}} \mathbb{E}[\Pi^W] = \max_{\{q_t^W, p_t^W\}_{t=1}^{24}} \sum_{t=1}^{24} \mathbb{E} \left[\lambda_t \cdot d_t^W - \kappa \cdot \max(0, d_t^W - \tilde{f}_t^W) \right] \quad (19)$$

subject to constraints (5) and (6). The expectation is taken jointly over $\tilde{D}_t$, $\tilde{f}_t^W$, $\tilde{p}_t^{C_k}$, and $\tilde{p}_t^S$.

The trade-off embedded in (19) operates along two dimensions. On bid price: a higher p_t^W reduces dispatch probability but raises λ_t when accepted, whereas a lower p_t^W increases dispatch probability but may depress λ_t if the wind producer is marginal. On bid quantity: a higher q_t^W increases potential revenue when dispatched but enlarges expected shortfall $\mathbb{E}[\Delta_t^W]$ and associated penalty costs; a lower q_t^W reduces imbalance risk but forgoes revenue. Because λ_t and d_t^W are non-linear, discontinuous functions of all bids and stochastic variables, (19) is evaluated by Monte Carlo simulation.

When Perspective W is evaluated in isolation, κ is treated as a fixed exogenous scalar.

Perspective R — Regulator Optimisation

Given the wind producer's best-response function $\mathbf{u}^W(\kappa)$ and distributional beliefs over all stochastic variables, the regulator solves:

$$\max_{\kappa \in [\underline{\kappa}, \bar{\kappa}]} \mathbb{E}[\mathcal{W} \mid \mathbf{u}^W(\kappa)] \quad (20)$$

subject to $\underline{\kappa} \leq \kappa \leq \bar{\kappa}$. The regulator's problem is a Stackelberg leadership problem: κ is set first and publicly announced; the wind producer then responds optimally. The regulator must therefore nest the wind producer's optimisation (19) within its own search over κ , evaluating expected welfare at each candidate penalty rate.

The optimal penalty κ^* balances the following forces: a penalty that is too low relative to c_t^{bal} leaves imbalance risk under-priced, inducing aggressive bidding and frequent large shortfalls; a penalty that is too high suppresses renewable dispatch and reduces both producer surplus and consumer surplus. The socially efficient penalty aligns the wind producer's private marginal imbalance cost with the system's marginal balancing cost.

When Perspective R is evaluated in isolation, $\mathbf{u}^W$ is treated as a fixed exogenous vector.

Composed (Stackelberg) Equilibrium

In the fully composed model, both optimisations are active simultaneously. The solution concept is a Stackelberg equilibrium $(\kappa^*, \mathbf{u}^{W*})$ satisfying:

$$\kappa^* = \arg \max_{\kappa} \mathbb{E}[\mathcal{W} \mid \mathbf{u}^W(\kappa)] \quad (21)$$

$$\mathbf{u}^{W*} = \mathbf{u}^W(\kappa^*) = \arg \max_{\mathbf{u}^W} \mathbb{E}[\Pi^W \mid \kappa^*] \quad (22)$$

This is solved by bi-level Monte Carlo optimisation: an outer loop searches over κ ; for each candidate κ , an inner loop solves the wind producer's problem (19) to obtain $\mathbf{u}^W(\kappa)$; the outer loop then evaluates $\mathbb{E}[\mathcal{W} \mid \mathbf{u}^W(\kappa)]$ and updates κ accordingly.

3.8 Key Modelling Assumptions

1. **Single-node network:** no transmission constraints; the market is copper-plate.
2. **Inelastic demand:** $\tilde{D}_t$ is exogenous and must be fully satisfied in every interval.
3. **Price-taking competitors:** conventional and solar producers follow fixed behavioural rules; they do not optimise and do not respond to κ .
4. **Strategic wind producer:** the wind producer is the sole strategic bidding agent in the market.

5. **Stackelberg structure:** the regulator is the Stackelberg leader and moves first by announcing κ ; the wind producer is the follower and responds optimally.
6. **Asymmetric imbalance settlement:** only under-delivery by the wind producer is penalised; over-production above the dispatch commitment is curtailed without compensation.
7. **Penalty as a transfer:** the penalty payment $\kappa \cdot \Delta_t^W$ is a transfer from the wind producer to the system and does not constitute a social loss; only C_t^{sys} enters the welfare calculation.
8. **No inter-interval linkages:** each interval is cleared independently; no ramping, storage, or carry-over constraints.
9. **Independence across intervals:** all stochastic variables are drawn i.i.d. across t .
10. **Monte Carlo evaluation:** all expectations in (19) and (20) are evaluated by Monte Carlo simulation owing to the non-linearity and discontinuity of λ_t and d_t^W as functions of bids and stochastic variables.

Now, please provide a modular Python implementation of the unified model, with each agent implemented as a separate class and a single interface function defined at the end.

All four modes run cleanly.

The implementation is ~650 lines of pure NumPy Python. Here is a summary of how the ODD specification maps onto the code.

Class structure

Class	ODD role	Key method(s)
<code>Environment</code>	Shared environment (Part I)	<code>draw_uncertain_parameters</code> , <code>draw_demand</code> , <code>draw_wind_production</code> , <code>solar_profile</code>
<code>ConventionalProducer</code>	Non-strategic competitor	<code>get_bids</code> — constant capacity, stochastic price (eq. 1-2)
<code>SolarProducer</code>	Non-strategic competitor	<code>get_bids</code> — cosine profile, stochastic price (eq. 3-4)
<code>WindProducer</code>	Perspective W focal agent	<code>set_bids</code> , <code>get_bids</code> , <code>settle</code> (eq. 14-16), <code>optimise_bids</code> (eq. 18-19)
<code>MarketOperator</code>	Passive infrastructure	<code>clear_interval</code> — two-stage merit-order + priority rationing (eq. 7-12)
<code>SystemRegulator</code>	Perspective R focal agent	<code>set_kappa</code> , <code>compute_welfare</code> (eq. 20-21), <code>optimise_kappa</code> (eq. 21-22)

Internal helpers

Two private functions keep the interface clean. `_collect_bids` assembles the full bid dictionary from all agents for one run. `_run_one_day` executes the 7-step daily schedule (stochastic draws → clearing → settlement → welfare accounting) and returns a `DayOutcome`.

`run_simulation` interface

The single public entry point handles all four evaluation modes via the `mode` argument. Both perspectives' controls (`kappa`, `q_W`, `p_W`) are either optimised or treated as fixed exogenous inputs depending on the mode, enabling fully modular evaluation as specified. The `'stackelberg'` mode implements the bi-level loop exactly as in eq. 21-22: the outer loop searches over `kappa`; the inner loop calls `wind_agent.optimise_bids(kappa)` to

obtain the best response; the outer loop then evaluates `E[W | u_W(kappa)]` and updates `kappa_star`.

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The wind-power producer's bid quantities should not be bounded by the mean of actual wind-power production. Please refine your implementation accordingly.

The change has three locations: the `Environment` class (add installed wind capacity $\bar{Q}^W$ as a constant), the `WindProducer` class (replace `mu_W_q` with `Q_W` as the upper bound on bid quantity), and the `run_simulation` interface (add the new parameter).

Now I have a complete picture. There are six targeted changes needed:

1. `Environment` — add `Q_wind` parameter and store as `self.Q_wind`
2. `WindProducer` class docstring — update the constraint from `[0, mu_W_q]` to `[0, Q_wind]`
3. `WindProducer.set_bids` — clip to `self.env.Q_wind`
4. `WindProducer.optimise_bids` — use `self.env.Q_wind` as grid upper bound and default
5. `run_simulation` — add `Q_wind` parameter and pass through to `Environment`
6. `__main__` — add `Q_wind` to the example parameter set

All four modes run correctly. The results also validate the economic logic of the revision: in `stackelberg` mode the regulator now sets $\kappa^* = \$60/\text{MWh}$ (non-trivial, because overbidding relative to μ_q^W is now possible), and the wind producer responds by pulling its optimal bid quantity back to 53.33 MWh — well below the installed capacity of 80 MWh but also below the old ceiling of $\mu_q^W = 50$ MWh, correctly reflecting the trade-off between increased revenue and penalty exposure across the wider feasible range.

Summary of changes

The refinement touched six locations, each for a distinct reason:

Location	Change	Rationale
<code>Environment</code> docstring	Added <code>Q_wind</code> parameter description	Documents the new constant clearly, distinguishing it from <code>mu_W_q</code>
<code>Environment.__init__</code> signature	Added <code>Q_wind: float</code> argument	Makes installed capacity an explicit, separately configurable input
<code>Environment.__init__</code> body	Stored <code>self.Q_wind = float(Q_wind)</code>	Makes the constant available to all agents via the shared environment
<code>WindProducer</code> docstring & <code>set_bids</code>	Constraint updated from <code>[0, mu_W_q]</code> to <code>[0, Q_wind]</code> ; clip updated accordingly	The producer may commit up to installed capacity; overbidding relative to expected output is now a legitimate strategy whose risk is priced by κ
<code>WindProducer.optimise_bids</code>	Grid upper bound and default initialisation changed from <code>mu_W_q</code> to <code>Q_wind</code>	The optimisation now searches the full feasible range $[0, \bar{Q}^W]$, allowing the model to find bid quantities above μ_q^W when the penalty is low enough

Location	Change	Rationale
<code>run_simulation</code>	Added <code>Q_wind</code> parameter; passed to <code>Environment</code> ; updated default <code>q_w_init</code> to <code>Q_wind</code> ; updated docstring	Exposes the new constant through the single public interface



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